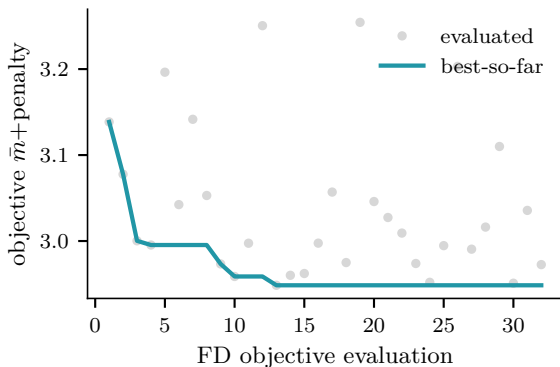
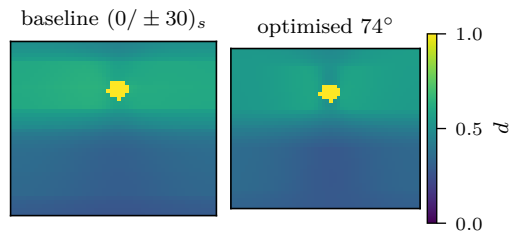


# Stage-5 design feedback: fleet-robust laminate optimisation (exact FD phase-field forward)

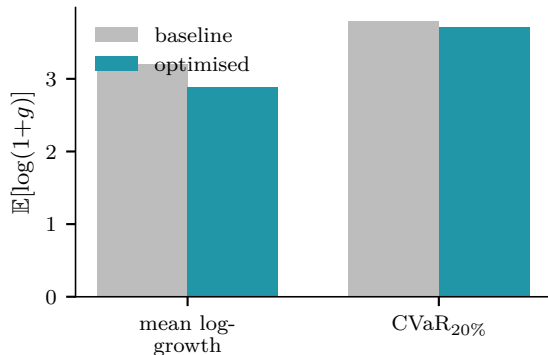
(a) Bayesian-optimisation convergence



(b) worst-case defect  $d_{\text{final}}$



(c) robust growth ( $-10\%$  mean,  $-2\%$  tail)



(d) objective vs off-axis angle

